



# Cambridge IGCSE™

## PHYSICS

Paper 1 Multiple Choice (Core)

0625/13

May/June 2026

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall =  $9.8 \text{ m/s}^2$ ).

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

- 1 A stone has an irregular shape.

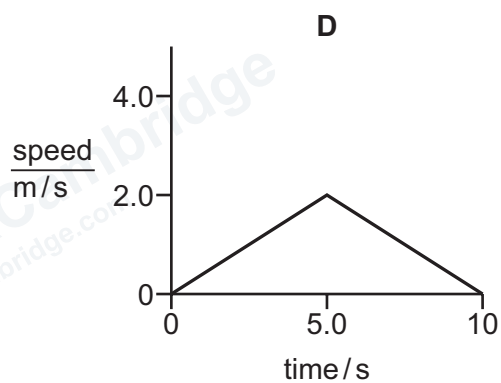
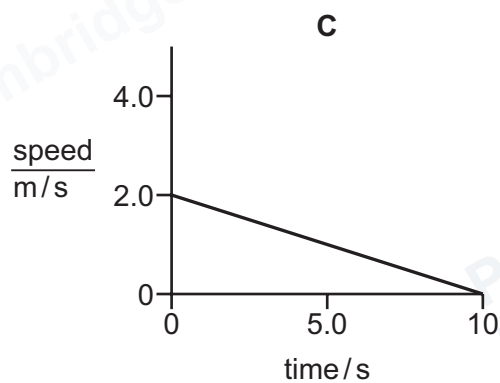
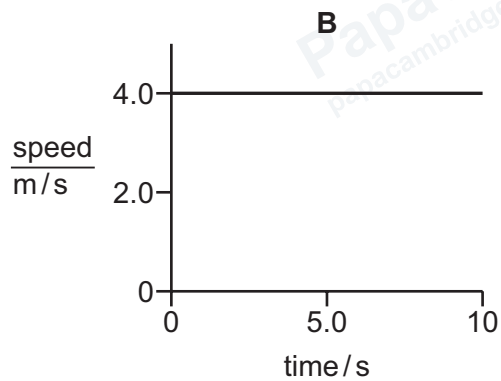
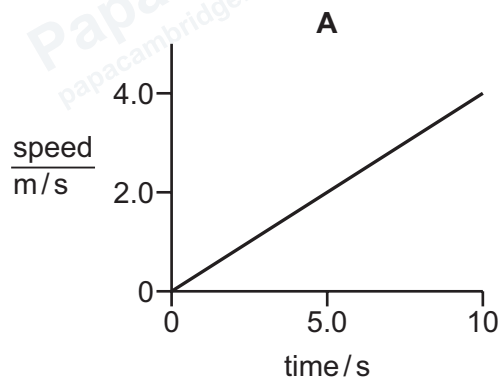
Which apparatus is needed to find its volume?

- A a thermometer
- B a measuring cylinder
- C a protractor
- D a ruler

- 2 What is the definition of velocity?

- A change in speed
- B distance travelled divided by the time taken
- C magnitude of the speed
- D speed in a given direction

- 3 Which speed–time graph is for an object that travels 20 m in 10 s?



- 4 Which row about weight and gravitational field strength is correct?

|          | weight is                                   | gravitational field strength is             |
|----------|---------------------------------------------|---------------------------------------------|
| <b>A</b> | equivalent to the acceleration of free fall | the gravitational force per unit mass       |
| <b>B</b> | equivalent to the acceleration of free fall | the gravitational force on an object        |
| <b>C</b> | the gravitational force per unit mass       | equivalent to the acceleration of free fall |
| <b>D</b> | the gravitational force on an object        | the gravitational force per unit mass       |

- 5 The table shows the densities of three liquids.

| liquid | density<br>g/cm <sup>3</sup> |
|--------|------------------------------|
| oil    | 0.8                          |
| water  | 1.0                          |
| brine  | 1.2                          |

A student has four balls with different masses and volumes.

Which ball would float on brine but sink in oil and in water?

|          | mass / g | volume / cm <sup>3</sup> |
|----------|----------|--------------------------|
| <b>A</b> | 26       | 20                       |
| <b>B</b> | 28       | 32                       |
| <b>C</b> | 30       | 38                       |
| <b>D</b> | 34       | 30                       |

- 6 A turning effect of 25 N m is needed to loosen a nut on a car engine. A mechanic uses a spanner of length 30 cm.

What is the minimum force the mechanic must apply to the spanner to loosen the nut?

- A** 1.2 N      **B** 7.5 N      **C** 30 N      **D** 83 N

7 What is the name of the point through which the weight of an object appears to act?

- A the centre of gravity
- B the moment
- C the pivot
- D the turning point

8 An electric car is travelling at **constant** velocity along a straight horizontal road.

Which row shows the energy changes that are taking place?

|   | energy store decreasing | transfer process | energy store increasing |
|---|-------------------------|------------------|-------------------------|
| A | chemical                | electrical       | thermal                 |
| B | chemical                | electrical       | kinetic                 |
| C | electrical              | mechanical       | kinetic                 |
| D | electrical              | mechanical       | thermal                 |

9 Which statement is correct?

- A Energy is the power transferred per unit time.
- B Energy is the time transferred per unit power.
- C Power is the energy transferred per unit time.
- D Power is the time transferred per unit energy.

10 Sea water is denser than fresh water.

Where is the pressure on a diver greatest?

- A 10 m below the surface of a fresh-water lake
- B 10 m below the surface of the sea
- C 20 m below the surface of a fresh-water lake
- D 20 m below the surface of the sea

- 11 The air pressure in a car tyre is measured on a warm day. The pressure is then measured again during the cold night. The air pressure in the tyre has decreased. There are no air leaks in the tyre.

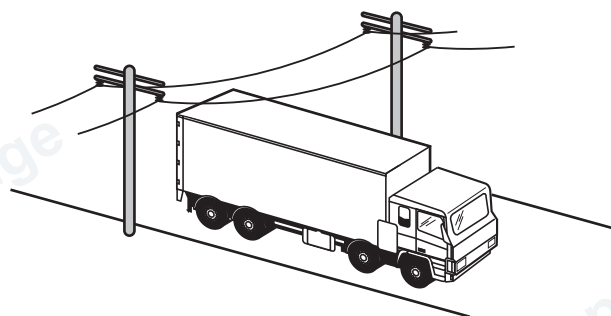
Why does the pressure decrease in the cold at night?

- A The air particles in the tyre move more slowly.
- B The air particles in the tyre stop moving.
- C The volume of the air in the tyre decreases.
- D The volume of the air in the tyre increases.

- 12 Which equation is used to convert temperatures between  $T$  in kelvin and  $\theta$  in degrees Celsius?

- A  $\theta = T + 273$     B  $\theta = T - 273$     C  $\theta = 273 - T$     D  $\theta = 273 \times T$

- 13 The diagram shows a lorry travelling under electricity transmission cables.



What is a possible hazard when the cables are too long?

- A In cold conditions, the cables contract and might be hit by the lorry.
- B In hot conditions, the cables expand and might be hit by the lorry.
- C In cold conditions, the cables expand and break.
- D In hot conditions, the cables contract and break.

- 14 Which statement is correct about the particles of a substance after condensation?

- A They are close to each other and slide over each other.
- B They are close to each other and vibrate about fixed points.
- C They are far apart from each other and vibrate about fixed points.
- D They are far apart from each other and move freely within the container.

- 15 At the surface of a liquid, the more-energetic molecules can escape from the liquid into the atmosphere.

Which name is given to this process?

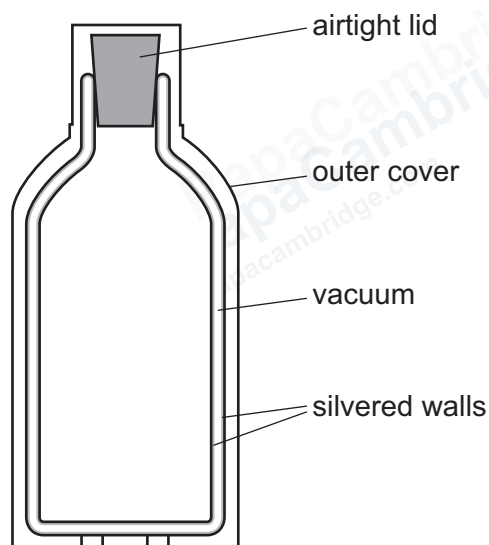
- A boiling
- B condensation
- C evaporation
- D melting

- 16 Convection is an important method of thermal energy transfer.

In which states of matter does convection occur?

- A gases, liquids and solids
- B gases and liquids only
- C gases and solids only
- D liquids and solids only

- 17 The diagram shows a vacuum flask used to keep a liquid warm.



Which methods of heat loss are reduced by the **vacuum** between the silvered walls?

- A conduction only
- B conduction and convection only
- C convection and radiation only
- D conduction, convection and radiation

**18** Waves transfer energy.

Which statement does **not** describe a wave?

- A** vibrations in the earth from an earthquake travelling to your feet
- B** light from a lamp travelling to your eye
- C** sound from a person singing travelling to your ear
- D** wind from a fan travelling to your face

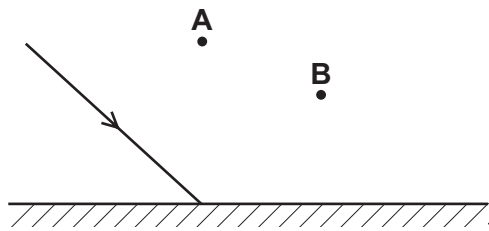
**19** The speed of sound in water is 1500 m/s.

What is the wavelength in water of a sound of frequency 26 kHz?

- A** 17 mm                      **B** 58 mm                      **C** 17 m                      **D** 58 m

**20** The diagram shows a ray of light striking a plane mirror.

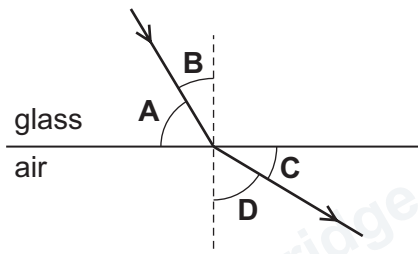
Through which point does the reflected ray pass?



- D**                      **C**

**21** A narrow beam of light travels through glass. It reaches the edge of the glass and refracts into the air.

What is the angle of refraction?



22 Which list states three colours of the visible spectrum in order of decreasing frequency?

- A red, yellow, blue
- B orange, green, blue
- C indigo, yellow, green
- D violet, orange, red

23 Which region of the electromagnetic spectrum is between visible light and X-rays?

- A gamma rays
- B infrared
- C radio
- D ultraviolet

24 Which region of the electromagnetic spectrum is used to detect fake bank notes?

- A microwaves
- B radio
- C ultraviolet
- D X-rays

25 An audible sound wave and an ultrasound wave are travelling in air.

The table shows information about properties of the waves.

Which property gives possible correct information for both the audible sound wave and the ultrasound wave?

|   | property   | audible sound | ultrasound |
|---|------------|---------------|------------|
| A | frequency  | 3.0 kHz       | 30 kHz     |
| B | speed      | 330 m/s       | 3300 m/s   |
| C | wavelength | 1.0 m         | 10 m       |
| D | wave type  | longitudinal  | transverse |



26 The hearing ranges of four animals are shown.

| animal   | lowest frequency / Hz | highest frequency / Hz |
|----------|-----------------------|------------------------|
| cow      | 25                    | 35 000                 |
| dog      | 67                    | 45 000                 |
| elephant | 17                    | 12 000                 |
| ferret   | 16                    | 44 000                 |

Which animals can hear both lower frequencies than a human and higher frequencies than a human?

- A cow and dog
- B dog only
- C elephant and ferret
- D ferret only

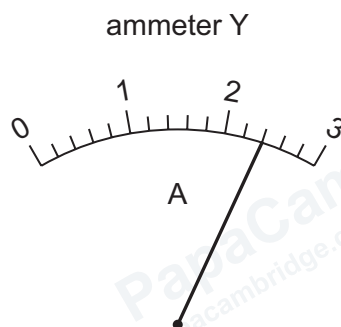
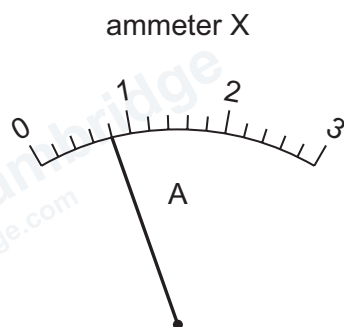
27 What is **not** an example of induced magnetism being used?

- A using a compass needle to find the direction of the Earth's magnetic field
- B magnetising a soft-iron bar by placing it in a strong magnetic field
- C using an electromagnet to pick up pieces of scrap iron
- D using a soft-iron core in a transformer

28 Which statement explains why copper is a good electrical conductor?

- A Copper is a non-metal.
- B Electrons can flow freely in copper.
- C Particles in copper are closely packed.
- D The density of copper is very low.

29 The readings on two ammeters are shown.



Which statement about the readings is correct?

- A** The current shown on ammeter Y is 2.4 times larger than the reading on ammeter X.
- B** The current shown on ammeter Y is 3.0 times larger than the reading on ammeter X.
- C** The potential difference (p.d.) shown on ammeter Y is 2.4 times larger than the reading on ammeter X.
- D** The potential difference (p.d.) shown on ammeter Y is 3.0 times larger than the reading on ammeter X.

30 A lamp rated 12 V, 24 W operates at normal brightness.

How much energy is transferred by the lamp in 20 s?

- A** 1.2 J                      **B** 240 J                      **C** 480 J                      **D** 5800 J

31 A length of metal wire is used as a resistor in a circuit.

Which change results in an increase in current in the circuit?

- A** connecting a second identical length of wire in parallel with the original wire
- B** connecting a second identical length of wire in series with the original wire
- C** decreasing the thickness of the wire
- D** increasing the length of the wire

**32** A teacher asks, 'Why do we put a fuse in a mains circuit?'

Student 1 says, 'It protects the wiring from overheating.'

Student 2 says, 'It protects us from getting a shock if we touch the live wire.'

Who is correct?

- A** both students
- B** neither student
- C** student 1 only
- D** student 2 only

**33** The circuits in a house are protected by trip switches.

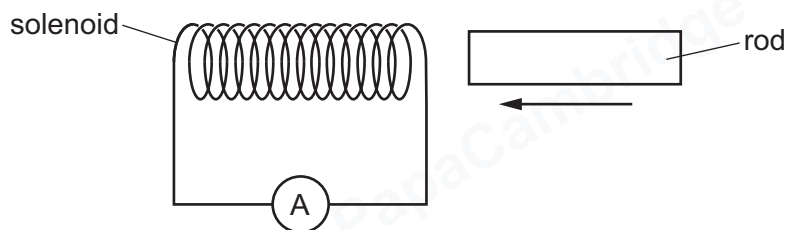
A trip switch is connected to an electric shower that uses 50 A.

Which trip switch rating is most suitable?

- A** 5.0 A
- B** 50 A
- C** 60 A
- D** 90 A

**34** A solenoid is connected to a very sensitive ammeter. A rod is inserted into one end of the solenoid.

The ammeter shows that there is a small electric current in the solenoid while the rod is moving.

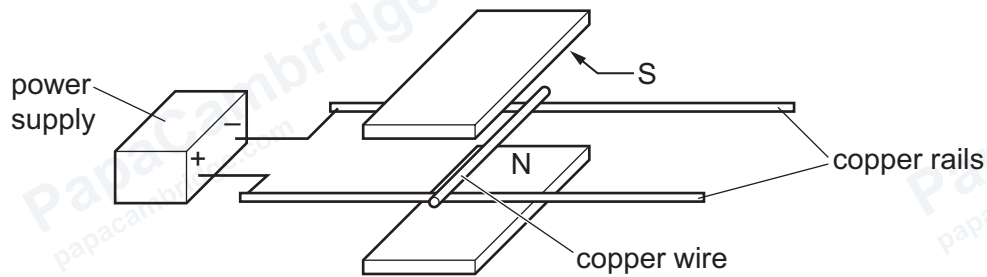


Which rod is being inserted?

- A** a heated copper rod
- B** a magnetised steel rod
- C** an uncharged nylon rod
- D** a radioactive uranium rod

- 35 The diagram shows two horizontal copper rails connected to a d.c. power supply.

A thin piece of copper wire rests on the rails in a vertical magnetic field.



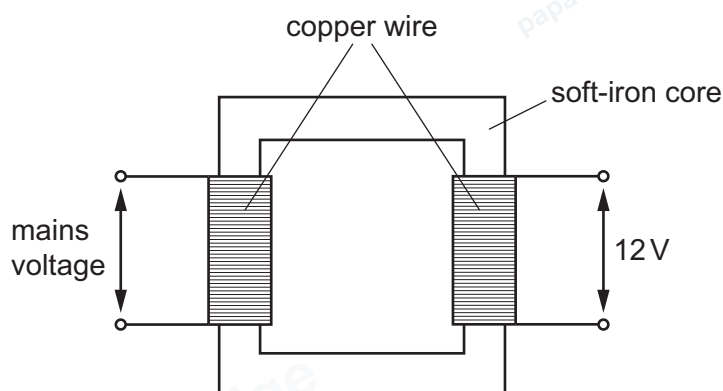
When the power supply is switched on, the thin piece of copper wire experiences a force to the right.

The magnetic field and the connections to the power supply are both reversed.

What now is the direction of the force on the thin piece of copper wire?

- A to the left
- B to the right
- C downwards
- D upwards

- 36 The step-down transformer shown reduces mains voltage to 12 V.



When the transformer is used, some energy is transferred to the surroundings.

Which energy store of the surroundings increases?

- A chemical
- B nuclear
- C thermal
- D elastic

37 How do atoms form positive ions?

- A by gaining electrons
- B by gaining neutrons
- C by losing electrons
- D by losing protons

38 Which statement is correct for an  $\alpha$ -particle?

- A It contains two neutrons.
- B It has a charge of  $-2$ .
- C It is **not** very ionising.
- D It is very penetrating.

39 Which statement about  $\alpha$ -particles and  $\gamma$ -rays is correct?

- A  $\alpha$ -particles are a form of electromagnetic radiation.
- B  $\alpha$ -particles penetrate materials more easily than  $\gamma$ -rays.
- C The emission of an  $\alpha$ -particle produces a nucleus of a different element.
- D  $\gamma$ -rays are more ionising than  $\alpha$ -particles.

40 The diameter of the Milky Way is approximately .....1..... .....2..... .

Which row correctly completes the sentence?

|   | 1           | 2           |
|---|-------------|-------------|
| A | 100 000     | kilometres  |
| B | 100 000 000 | kilometres  |
| C | 100 000     | light-years |
| D | 100 000 000 | light-years |



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